

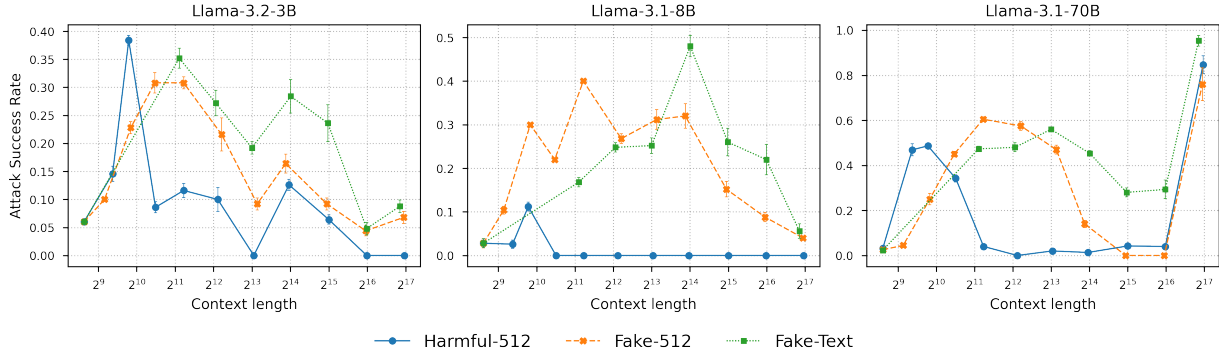


Figure 8: **ASR Comparison with Fake Data on Llama Models.** *Fake-512* and *Fake-Text* demonstrate comparable or even higher ASR than *Harmful-512*.

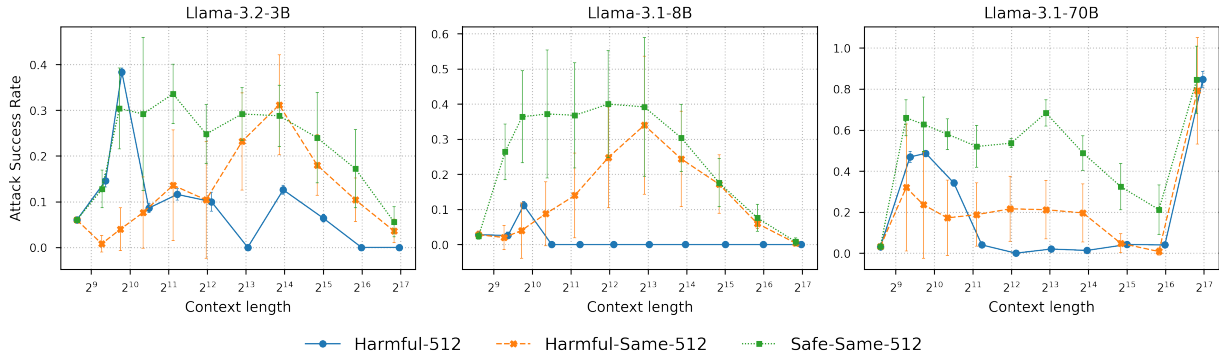


Figure 9: **Impact of Repeated Examples on Llama Models.** Repeating the same shot multiple times (*Harmful-Same-512* and *Safe-Same-512*) leads to consistently higher ASR compared to *Harmful-512*.

Figure 8 contrasts the performance of *Harmful-512* data against *Fake-512* and *Fake-Text* data, with Llama-3.1 models showing notably higher ASR when using fake content. These results indicate that even well-aligned models become vulnerable to jailbreaking attempts when random text fills the weakness points identified in our analysis. Combined with our findings from Section 4.3 where safe content showed superior effectiveness to harmful content, these results highlight fundamental limitations of input-based safety filters. Moreover, the higher vulnerability of larger models to these attacks reinforces the concerning trade-off between model capabilities and safety.

5.2 Shot Repetition Attack

Another approach to fill context is simply repeating the same shot multiple times. Unlike fake content which cannot utilize harmfulness characteristics, this repetition strategy further exploits the previously observed vulnerability where Llama models showed higher susceptibility to safe content. This is highly efficient as it requires collecting only one set of QA pairs and repeating them to fill the context. To evaluate the effectiveness of this strategy, we

conduct experiments using *Harmful-512*, *Harmful-Same-512*, and *Safe-Same-512* datasets.

As shown in Figure 9, the Llama models exhibit particularly high vulnerability to safe shots, consistent with our previous findings. Detailed examples of the QA pairs used in *Harmful-Same-512* and *Safe-Same-512* datasets are provided in Appendix G. These results indicate that effective jailbreaking attacks can be achieved by simply repeating safe examples, without requiring complex example generation or careful content selection. The consistent effectiveness of these simplified attack strategies suggests that future safety research may need to consider how models process repeated content in context length.

6 Analysis of Vulnerabilities and Defense Strategies

Our findings on vulnerability pattern with performance degradation align with fundamental characteristics in long-context interactions. Li et al. (2023a) found that models handling extended contexts often show decreased performance compared to shorter sequences, which aligns with the degra-

dation phase observed in our experiments. Furthermore, Liu et al. (2024a) identified that models exhibit diminished capability in middle portions, while maintaining stronger performance at sequence boundaries. These insights emphasize how increased context length negatively impacts safety alignment in many-shot scenarios.

Based on these findings, it is crucial to examine existing defense approaches. Jailbreak defense strategies can be categorized into prompt-level and model-level approaches. Prompt-level defenses include prompt detection (Alon and Kamfonas, 2023; Jain et al., 2023), perturbations (Cao et al., 2024; Zhou et al., 2024a), and system prompt safeguards (Sharma et al., 2024; Wang et al., 2024a; Zheng et al., 2024a). These approaches may be vulnerable to Many-Shot attacks, as shown in Figures 2, 6, and 9, succeeding even with harmless instructions or non-malicious shots.

At the model level, Supervised Fine Tuning (SFT) (Deng et al., 2023; Bhardwaj and Poria, 2023; Bianchi et al., 2024) and Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al., 2022; Dai et al., 2024; Siththarajan et al., 2024) have shown promise. Both Llama (AI@Meta, 2024a,b) and Qwen (Yang et al., 2024; Team, 2024) have enhanced their safety, particularly with Llama 3.1 reporting MSJ defense improvements through SFT (Llama Team, 2024). However, vulnerabilities persist across models, raising questions about the effectiveness of current training procedures.

To understand how training procedures impact these vulnerabilities, we analyzed Llama-3.1-8B models before instruction tuning. As shown in Figure 10, the base model demonstrated expected behavior with increased vulnerability to Harmful-512 attacks while maintaining robust defense against safe content. However, instruction-tuned models appear to exhibit different patterns: while defense against Harmful-512 attacks improved, vulnerability to Safe-512 attacks increased, suggesting an inverse relationship that warrants further investigation into current safety training procedures.

Given these observations, current defense strategies may not adequately address the unique challenges of long-context interactions. Our results suggest that safety alignment varies across different phases of extended conversations, indicating the need for defense mechanisms specifically designed for long-context scenarios.

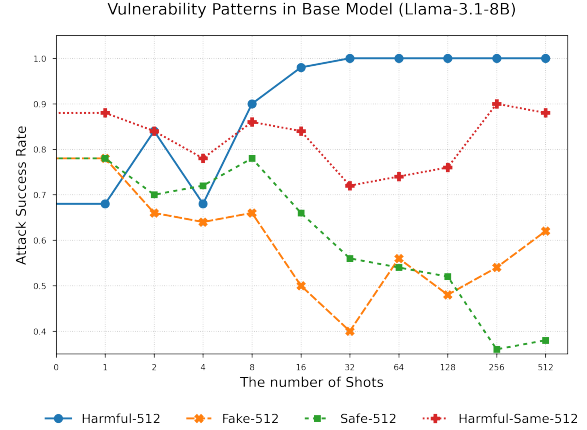


Figure 10: **Base Model ASR Analysis.** Llama-3.1-8B Base model shows expected vulnerability patterns with higher ASR for harmful content.

7 Related Work

Scaling In-Context Learning for Long Contexts

Increasing the number of examples in In-Context Learning enhances task performance across language tasks (Brown et al., 2020; Lu et al., 2022). To leverage this advantage at scale, recent innovations in attention mechanisms and context processing have enabled efficient handling of longer contexts (Zheng et al., 2023; Ding et al., 2024; Team et al., 2024; Xiong et al., 2024). The effectiveness of ICL depends heavily on example selection strategies, with demonstration diversity and relevance playing crucial roles across different shot settings (Zhao et al., 2021; Lu et al., 2022; Liu et al., 2022; Levy et al., 2023; Wan et al., 2023; Zhou et al., 2024b; Wan et al., 2024; Anonymous, 2024).

Recent work has significantly expanded the boundaries of ICL by increasing both the number of examples and the context length: Li et al. (2023b) scaled up to 2K shots, Bertsch et al. (2024) extended context windows to 80K tokens, and Agarwal et al. (2024); Jiang et al. (2024) pushed these limits further with up to 1M tokens, demonstrating consistent performance improvements in long-context settings.

Jailbreaking and Safety in Many-Shot ICL

Recent studies have revealed critical vulnerabilities in advanced LLMs through topic-based attacks (Deshpande et al., 2023; Liu et al., 2024c) and sophisticated attack techniques (Zou et al., 2023; Chao et al., 2023; Liu et al., 2024b), leading to new defense mechanisms (Bai et al., 2022; Inan et al., 2023; Jain et al., 2024). In addition to these devel-

opments, few-shot jailbreak attempts (Rao et al., 2024; Wei et al., 2024; Zheng et al., 2024b) have emerged alongside ICL advancements.

Most notably, Many-Shot Jailbreaking (Anil et al., 2024) reveals that both attack effectiveness and general ICL capabilities follow power law patterns in long-context settings. Motivated by these advances, our work investigates the implications of extended context lengths on model safety, focusing on vulnerabilities in many-shot scenarios.

8 Conclusion

We present a comprehensive analysis of LLMs vulnerabilities across different context lengths, investigating vulnerability patterns across multiple model architectures in Many-Shot attack settings. Our experiments reveal several crucial findings. Model vulnerabilities are determined by model properties rather than attack characteristics, with context length being the primary influencing factor. Even well-aligned models show unexpected susceptibility, with attacks succeeding using meaningless text or simple repeated examples, suggesting current alignment strategies may be inadequate.

These discoveries highlight fundamental limitations inherent to model architectures in long-context processing capabilities. Our research demonstrates the need for safety mechanisms beyond current input-based defense strategies. Maintaining safety alignment across context lengths, improving context processing mechanisms, and developing position-aware safety mechanisms remain important directions for future research.

Limitations

Model Accessibility and Coverage Limitations

Our study focused on open-source models (e.g., Llama, Qwen) accessible under permissible licenses and usage policies, but we could not extend our experiments to proprietary models due to restrictive conditions. Transparency remains limited, as alignment tuning methodologies and training recipes are often undisclosed even for open-source models. Increased transparency in alignment methodologies would contribute to advancing the development of safer AI systems.

Evaluation Methodology Limitations We employed GPT-4o as the sole judge, which, despite high accuracy, may introduce evaluation biases. The interpretation of harmfulness can vary by observer, influenced by cultural and contextual factors.

Additionally, the datasets and queries we collected may not represent the full spectrum of harmful content. Our binary "safe/unsafe" classification oversimplifies complex behavioral patterns and may fail to fully capture subtle forms of potential harm.

Mechanistic Understanding Limitations While we highlight long-context vulnerabilities, our mechanistic understanding remains limited. Further examination of context expansion techniques, attention mechanisms, and architectural factors could clarify the origins of these observed behaviors. Future studies employing attention weight analysis, hidden state visualization, and embedding comparisons may offer deeper insights into the underlying processes driving model vulnerabilities.

Ethical Considerations

This research investigates critical vulnerabilities in long-context processing of large language models. While identifying these vulnerabilities is crucial for developing safer AI systems, we recognize the potential risks of our findings being misused. Our experiments utilize publicly available jailbreaking prompts for reproducibility and transparency, avoiding the introduction of novel harmful content. To ensure responsible research conduct, we carefully curated our experimental datasets and handled all collected data in accordance with usage policies and license.

All experimental findings and code will be shared under strict access controls with verified researchers. Examples of harmful content used in this paper are replaced with placeholders to prevent misuse while maintaining scientific validity. We conducted our research on publicly available models without legal restrictions, ensuring our methodology aligns with responsible disclosure practices in security research. Our findings aim to contribute to ongoing safety initiatives in the AI community by highlighting critical areas for improvement in long-context processing capabilities.

Acknowledgments

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